C. LIS or Reverse LIS?

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: constructive algorithms, greedy, implementation, math | https://codeforces.com/problemset/problem/1682/C

You are given an array a of n positive integers.

Let LIS(a) denote the length of longest strictly increasing subsequence of a. For example,

- LIS([2, underline1, 1, underline3]) = 2.
- LIS([underline3, underline5, underline10, underline20]) = 4.
- LIS([3, underline1, underline2, underline4]) = 3.

We define array a' as the array obtained after reversing the array a i.e. $a' = [a_n, a_{n-1}, \ldots, a_1]$.

The beauty of array a is defined as min(LIS(a),LIS(a')).

Your task is to determine the maximum possible beauty of the array a if you can rearrange the array a arbitrarily.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the length of array a.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the elements of the array a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum possible beauty of a after rearranging its elements arbitrarily.

Example

Input
3 3 6 6 6 6 2 5 4 5 2 4 4 1 3 2 2

Output
1 3 2

Note

In the first test case, a = [6, 6, 6] and a' = [6, 6, 6]. LIS(a) = LIS(a') = 1. Hence the beauty is $min(1, 1) = 1$.

In the second test case, a can be rearranged to [2, 5, 4, 5, 4, 2]. Then a' = [2, 4, 5, 4, 5, 2]. LIS(a) = LIS(a') = 3. Hence the beauty is 3 and it can be shown that this is the maximum possible beauty.

In the third test case, a can be rearranged to [1, 2, 3, 2]. Then a' = [2, 3, 2, 1]. LIS(a) = 3, LIS(a') = 2. Hence the beauty is $min(3, 2) = 2$ and it can be shown that 2 is the maximum possible beauty.